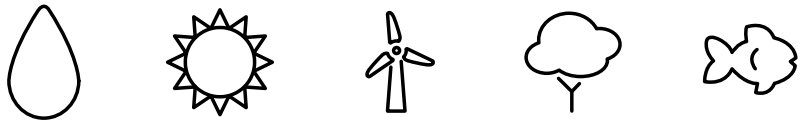
EARTH AND SPACE

Earth's Resources and Sustainability

Resources that cycle and resources that run out, technology, conservation and how science helps



droplet sun wind energy deciduous tree fish

In this pack

Learn it: Earth's resources: use them wisely ★☆☆
Fact check: Renewable or finite? ★☆☆
Think it through: Help or hindrance? ★★☆
Investigate: How much water does a dripping tap waste? ★★☆
Real-world case study: Catching the rain on the coast ★★★
Review and challenge: Show what you know ★★★

This pack belongs to: _____ **Class:** _____

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LEARN IT · B4 · RESOURCES

Earth's resources: use them wisely

★☆☆
Grade 4

Name: Class: Date:

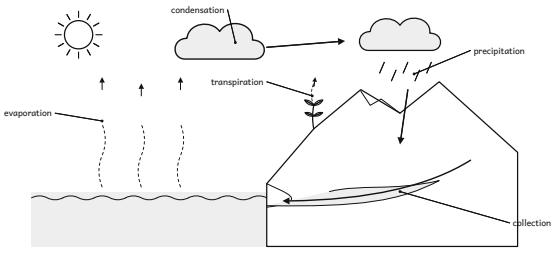
Learning goals

- Explain that some resources cycle through the environment while others will run out.
- Describe how technology can help and harm sustainability.
- Describe ways people conserve and protect resources, and how science helps.

Everything we use comes from the Earth: the water we drink, the rice we eat, the wood in our furniture, the gas in our stoves and the metal in our phones. These are **natural resources**. Some will last for ever if we look after them. Others will one day run out.

Resources that cycle

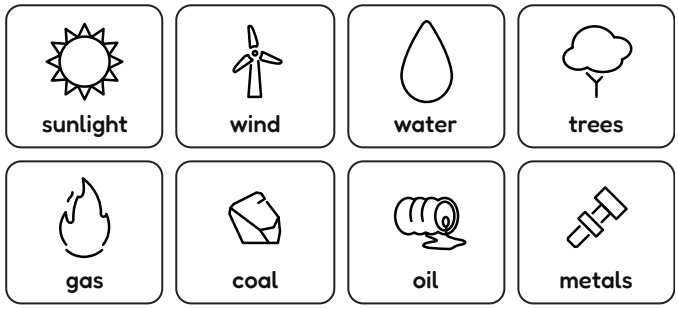
Some resources are used and then return to the environment to be used again. **Water** goes round and round in the **water cycle**: it evaporates from rivers and seas, forms clouds, falls as rain and flows back to the sea. **Soil** nutrients cycle when dead plants and animals rot. **Air** is cleaned by plants. Resources that are replaced naturally are **renewable**: sunlight, wind, water, trees and fish, if we do not use them faster than they are replaced.



Water is a resource that cycles.

Resources that run out

Other resources took millions of years to form and cannot be replaced. They are **finite** or **non-renewable**: **natural gas**, **coal**, **oil** and **metals** such as iron and copper. When they are used up, they are gone.



Bangladesh connection

Most of Bangladesh's electricity has been made by burning **natural gas** from fields under the ground. These gas fields are slowly running out, so the country is building more **solar** power and buying fuel from abroad.

Technology: help or harm?

Technology	How it helps	How it can harm
tube-wells	safe drinking water instead of dirty ponds	some water has arsenic; too much pumping lowers the water underground
plastic	cheap, light and waterproof	lasts hundreds of years as litter
motor boats and cars	fast travel and trade	burn fuel and pollute the air
solar panels	clean electricity from sunlight	old panels and batteries must be recycled safely

Looking after resources

- **Saving water**: turn off taps, fix leaks, collect rainwater.
- **Saving energy**: switch off lights and fans, use efficient LED bulbs and better cookstoves.
- **Protecting fish**: stop fishing while fish lay eggs (like the ban on catching hilsa).
- **Planting and protecting forests**: social forestry, where communities plant and care for trees.
- **Reduce, reuse, recycle** materials such as paper, metal and plastic.

Science helps solve problems

Scientists invent **test kits** to find arsenic in tube-well water, breed **rice that grows in salty water**, design **cleaner cookstoves** that use less wood and make less smoke, and measure pollution so that we know where to act.

Sudden and gradual changes

A **sudden** change, such as a cyclone, can destroy forests and fields in a day. A **gradual** change, such as the water underground slowly getting lower each year, is harder to notice but just as important. Both can upset the balance (stability) of an environment, so scientists measure resources every year.

Key words

natural resource	something from the Earth that people use
renewable	replaced naturally; will not run out if used carefully
finite (non-renewable)	will run out one day
conserve	use carefully so it lasts
sustainable	meeting our needs without spoiling things for the future
fossil fuel	coal, oil or gas formed from ancient living things
groundwater	water stored in the ground under our feet

FACT CHECK · B4 · RESOURCES

Renewable or finite?

★☆☆ Grade 4

Name: Class: Date:

Learning goals

- Sort resources.
- Name ways to conserve resources.

1 Sort the resources. [8]

SORT THESE

oil fish wind sunlight iron trees coal natural gas

Renewable (can be replaced)	Finite (will run out)

2 Match the action to the resource it saves. [4]

Draw a line to match each pair, or write the letter.

fixing a dripping tap ●	● **A** water
switching off the fan ●	● **B** fish
not catching young hilsa ●	● **C** electricity
using a cleaner cookstove ●	● **D** wood

3 Fill in the gaps. [3]

WORD BANK

wasting finite cycles conserving evaporating

Water is a resource that _____ through the environment. Gas and coal are _____: they will run out. Using resources carefully so they last is called _____ them.

Help or hindrance?

★★☆
Grade 4

Name: Class: Date:

Learning goals

- Evaluate technologies.
- Explain sustainable choices.

1 **Tube-wells gave millions of people in Bangladesh safe water, but some wells have arsenic. Is the tube-well a help or a hindrance? Explain both sides.** [3]

2 **Why is it a problem if people catch fish faster than the fish can breed?** [2]

3 **Natural gas will run out. Suggest two other ways Bangladesh could make electricity. Are they renewable?** [2]

4 **Give an example of a sudden change and a gradual change to a resource.** [2]

How much water does a dripping tap waste?

★★☆ Grade 4

Name: Class: Date:

Learning goals

- Measure volume and time.
- Use multiplication to scale up.
- Suggest action.

Class 4C put a measuring jug under a dripping tap for 10 minutes, three times.

1 **Complete the table.** [2]

Test	1	2	3	Mean
Water collected in 10 minutes (ml)	48	52	50	

2 **Use the mean to work out the water wasted.** [3]

Time	Water wasted
1 hour (6 × 10 minutes)	
1 day (24 hours)	
1 week (7 days)	

3 **Why did they repeat the test three times?** [1]

4 **A family needs about 3 litres of drinking water each day for each person. How many people could drink the water wasted in one day?** [2]

5 **Write a message for your school about saving water.** [2]

REAL-WORLD CASE STUDY · B4 · RESOURCES

Catching the rain on the coast

★★★ Grade 4

Name: Class: Date:

Learning goals

- Understand a real water problem.
- Calculate rainwater from a roof.
- Evaluate a sustainable solution.

Science | Maths | Health | Design

In the south-west coast of Bangladesh, in places like Satkhira and Khulna, the water in ponds and under the ground has become **salty**. Sea water floods in during cyclones and moves into the ground. Drinking salty water is bad for health, so many women and girls walk long distances each day to fetch fresh water.

One solution is **rainwater harvesting**. Rain falls on a roof, runs into a gutter and pipe, and is stored in a big clean tank. The monsoon brings plenty of rain, so a family can store water to drink during the dry season. Rain is a **renewable** resource that cycles through the environment.

Based on information about water salinity and rainwater harvesting in coastal Bangladesh.

1 **Maths. 1 mm of rain falling on 1 square metre of roof gives 1 litre of water.** [3]

Roof area	Rain in one storm	Water collected
20 square metres	10 mm	
20 square metres	50 mm	
40 square metres	50 mm	

2 **Why is rainwater safer to drink than the pond water on the coast?** [2]

.....

.....

3 **How does a rainwater tank help girls and women?** [2]

.....

.....

4 **Design a rainwater harvesting system for your school. Label the roof, gutter, pipe, filter and tank.** [3]

Rainwater system

Show what you know

★★★
Grade 4

Name: Class: Date:

Learning goals

- Use key vocabulary.
- Answer test-style questions.

1 Find the key words.

T	A	W	R	A	I	N	L	D	I
S	V	A	R	E	C	Y	C	L	E
G	L	R	E	S	O	U	R	C	E
A	J	F	I	S	H	W	E	R	J
S	J	F	M	T	R	A	F	B	H
F	O	R	E	S	T	T	K	N	C
R	B	E	E	A	D	E	G	L	H
M	C	O	N	S	E	R	V	E	T
C	V	H	O	V	K	R	V	G	B
Y	L	M	S	O	L	A	R	D	V

conserve fish forest gas rain recycle resource solar water

2 Which resource is finite? [1]

- ◯ sunlight
- ◯ wind
- ◯ natural gas
- ◯ rain

3 Which is a sustainable way to fish? [1]

- ◯ use nets with tiny holes
- ◯ catch fish all year
- ◯ stop fishing while fish lay eggs
- ◯ use poison

4 Draw the water cycle and label: evaporation, condensation, rain, rivers. [4]

Water cycle

5 **Challenge. "Trees are renewable, so we can cut down as many as we like." Do you agree? Explain.** [2]

.....

.....

.....

How am I doing? Colour or tick a circle.

I can...	Not yet	Nearly	Got it!
sort renewable and finite resources	○	○	○
explain how water cycles	○	○	○
say how a technology helps and harms	○	○	○
calculate water wasted by a dripping tap	○	○	○
describe ways to conserve resources	○	○	○
explain rainwater harvesting	○	○	○